General Instructions for Students

1. Use this format for every Applied Machine Learning laboratory experiment unless the faculty specifies otherwise.
2. Record the problem formulation and the machine-learning task clearly before writing code.
3. Document the dataset, preprocessing operations, model configuration, training procedure, evaluation metrics, and interpretation of results.
4. Write algorithms and pseudocode in your own words; do not merely paste source code.
5. Insert only relevant outputs, plots, tables, and screenshots. Every output should be interpreted.
6. For comparative experiments, report the same evaluation metrics for all models.
7. Mention random state, train-test split, hyperparameters, and any preprocessing choices required for reproducibility.
8. Conclude each experiment with observations, limitations, and the final result.
9. Complete viva-voce and faculty assessment sections after the experiment.

EXPERIMENT No. 1

Date of Experiment	19/08/2026	**Date of Submission**	19/08/2026
Experiment Title	Predicting Housing Prices Using Linear Regression	**Batch / Section**	8
Student Name	Abhijeet Panwar	**SAP ID**	590026303

1. Title of the Experiment

Write a concise title that clearly identifies the machine-learning task or algorithm being implemented.

Predicting Housing Prices Using Linear Regression

A regression experiment to estimate house prices from property and location-related features.

2. Problem Specification

State the problem to be solved. Identify the input features, expected output/target, type of ML task (classification, regression, clustering, etc.), and the required outcome.

Develop a supervised regression model to predict the selling price of a house using real-world property features such as area, bedrooms, bathrooms, number of stories, road access, guest room, basement, air conditioning, parking, preferred-area status, and furnishing status. The expected output is the continuous target variable price. The required outcome is a trained and evaluated Multiple Linear Regression model for house-price prediction.

3. Objectives

List the specific conceptual and implementation objectives of the experiment.

1. Understand house-price prediction as a supervised machine-learning regression task using a real Kaggle dataset.

2. Load, inspect, clean, and preprocess numerical and categorical housing features.

3. Convert categorical attributes into numerical form using one-hot encoding and train a Linear Regression model using an 80:20 train-test split.

4. Evaluate predictions using MAE, MSE, RMSE, and R^2 score.
5. Interpret feature relationships and visualize actual versus predicted house prices.

4. Learning Outcomes

State what a student should be able to understand, implement, evaluate, or interpret after completing the experiment.

After this experiment, the student will be able to:

• formulate a real-world regression problem for house-price prediction;
• preprocess mixed numerical and categorical data using a scikit-learn Pipeline;
• train and evaluate a Multiple Linear Regression model using MAE, MSE, RMSE, and R^2;

5. Prerequisite Concepts

Mention mathematical, statistical, programming, or ML concepts required before performing the experiment.

Basic Python programming and data handling with NumPy and Pandas.
Concepts of supervised learning, regression, independent variables, target variable, and train-test split.
Basic statistics: mean, variance, correlation, residual error, and coefficient of determination.
Understanding of Multiple Linear Regression and categorical-feature encoding.

6. Software / Hardware Requirements

Category	Requirement / Version
Programming Language	Python 3.x
IDE / Environment	Jupyter Notebook / VS Code
Libraries / Frameworks	NumPy, Pandas, Matplotlib, scikit-learn
Operating System	Windows / Linux / macOS
Hardware / Accelerator, if any	CPU; GPU not required

7. Dataset Description

Describe the dataset sufficiently so that another student can understand what data were used and how the learning task is defined.

Dataset Attribute	Details
Dataset Name	Housing Prices Dataset
Dataset Source / URL	https://www.kaggle.com/datasets/yasserh/housing-prices-dataset
Number of Samples	545
Number of Features	12 input features
Target Variable	price
ML Task	Supervised Regression

Dataset source selected for this experiment: Kaggle — Housing Prices Dataset by M Yasser H. File used: Housing.csv.

Feature Description

Feature / Attribute	Data Type	Role	Description
area	Integer	Input	Total area of the house in square feet
bedrooms	Integer	Input	Number of bedrooms

bathrooms	Integer	Input	Number of bathrooms
stories	Integer	Input	Number of stories
mainroad	Categorical	Input	Whether the house is connected to the main road (yes/no)
guestroom	Categorical	Input	Availability of a guest room (yes/no)

8. Theory

Explain the underlying concept or algorithm. Include important definitions, mathematical formulation, assumptions, working principle, advantages, and limitations wherever applicable.

Linear Regression is a supervised learning algorithm used to predict a continuous numerical target. In multiple linear regression, the target is modeled as a linear combination of several predictors: $y = \beta0 + \beta1x1 + \beta2x2 + ... + \beta nxn + \varepsilon$. During training, the model estimates coefficients that minimize the sum of squared residuals between actual and predicted values. In this experiment, house price is the target and property characteristics are predictors. Numerical features can be used directly, while categorical features must first be encoded into numeric columns. One-hot encoding is used here to avoid imposing an artificial numeric order on categories such as furnishing status. Performance is evaluated using MAE, MSE, RMSE, and R^2. Advantages of Linear Regression include simplicity, fast training, and interpretability; limitations include sensitivity to multicollinearity, outliers, omitted variables, and non-linear relationships.

9. Data Preprocessing

Record all preprocessing operations. Mark only those that are applicable and briefly justify each choice.

Preprocessing Step	Performed?	Method / Setting	Reason / Observation
Missing-value handling	No	No missing values found	All 545 records are complete
Duplicate removal	Yes	Checked using duplicated()	Remove repeated records if found
Categorical encoding	Yes	OneHotEncoder(drop="first")	Linear Regression requires numeric input
Feature scaling	No	Not required for ordinary Linear Regression	Original units retained for interpretability
Outlier treatment	No	No automatic deletion	Outliers should be interpreted rather than

			blindly removed
Feature selection	No	All available predictors used	Experiment uses the complete feature set
Data transformation	No	No log transformation in baseline model	Keeps the baseline experiment simple and reproducible

10. Exploratory Data Analysis (EDA)

Include relevant summary statistics, data distributions, correlations, class balance, and plots. Record the key observations below.

• The Kaggle Housing Prices Dataset contains 545 records and 13 columns: 12 input features and the target variable price.

• The data contains both numerical features (area, bedrooms, bathrooms, stories, parking) and categorical yes/no or furnishing-status features.

• The target price ranges from 1,750,000 to 13,300,000, while the mean price is approximately 4,766,729.

• The dataset contains no missing values in the 13 recorded columns. Categorical attributes require encoding before model training.

• The relationship between price and area is generally positive, but other features and multicollinearity also influence the final selling price.

11. Algorithm

Write the experiment-specific algorithm as clear numbered steps.

Step 1: Import the required Python libraries.

Step 2: Download the Housing.csv dataset from Kaggle and load it into a Pandas DataFrame.

Step 3: Inspect dataset shape, data types, missing values, duplicates, and summary statistics.

Step 4: Separate X as the 12 input features and y as the target variable price.

Step 5: Split the data into 80% training and 20% testing sets using random_state = 42.

Step 6: Apply OneHotEncoder to categorical columns and pass numerical columns unchanged.

Step 7: Initialize the preprocessing-and-Linear-Regression pipeline.

Step 8: Train the model on the training data and predict house prices for the testing data.

Step 9: Compute MAE, MSE, RMSE, and R^2 score.

Step 10: Visualize actual versus predicted prices and interpret the prediction errors.

12. Pseudocode

Write structured pseudocode for the complete ML workflow.

```
BEGIN
  Import Pandas, NumPy, Matplotlib, and scikit-learn libraries
  Load Housing.csv downloaded from the Kaggle dataset
  Inspect shape, data types, missing values, duplicates, and summary statistics
  Set X = all columns except price and y = price
  Identify numerical and categorical columns
  Apply one-hot encoding to categorical columns using a ColumnTransformer
  Split data into training (80%) and testing (20%) sets with random_state = 42
  Build a Pipeline containing preprocessing and Linear Regression
```

Fit the pipeline on the training data
Predict prices for testing data
Compute MAE, MSE, RMSE, and R^2
Plot actual versus predicted prices
Display sample predictions and interpret performance
END

13. Experimental Setup / Model Configuration

Record the parameters required to reproduce the experiment. Fill only the fields applicable to the chosen model.

Parameter	Value
Algorithm / Model	Linear Regression (sklearn.linear_model.LinearRegression)
Training Set (%)	80
Testing Set (%)	20
Validation Set (%), if used	Not used
Random State	42
Learning Rate	Not applicable
Epochs / Iterations	Not applicable
Batch Size	Not applicable
Regularization	None; baseline model
Other Hyperparameters	Categorical features encoded with OneHotEncoder(handle_unknown="ignore", drop="first")

14. Implementation / Source Code

Insert the complete, well-commented source code. Organize it into logical stages: imports, loading, preprocessing, model definition, training, prediction, evaluation, and visualization.

```
# Experiment 1: Predicting Housing Prices Using Linear Regression
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# 1. Load the Kaggle Housing Prices dataset
# Download Housing.csv from the Kaggle URL given in the dataset description
df = pd.read_csv("Housing.csv")

# 2. Basic data inspection
print("Dataset shape:", df.shape)
print("\nMissing values:\n", df.isnull().sum())
print("\nDuplicate rows:", df.duplicated().sum())
print("\nSummary statistics:\n", df.describe())

# 3. Separate input features and target
X = df.drop(columns="price")
y = df["price"]

# 4. Identify numerical and categorical columns
numerical_features = X.select_dtypes(include=["int64", "float64"]).columns.tolist()
categorical_features = X.select_dtypes(include=["object"]).columns.tolist()

# 5. Preprocessing: one-hot encode categorical features
preprocessor = ColumnTransformer(
    transformers=[
        ("num", "passthrough", numerical_features),
```

```
        ("cat", OneHotEncoder(handle_unknown="ignore", drop="first"), categorical_features)
    ]
)

# 6. Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42
)

# 7. Build and train the model pipeline
model = Pipeline(steps=[
    ("preprocessor", preprocessor),
    ("regressor", LinearRegression())
])
model.fit(X_train, y_train)

# 8. Generate predictions
y_pred = model.predict(X_test)

# 9. Evaluate the model
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print(f"\nMAE  : {mae:,.2f}")
print(f"MSE  : {mse:,.2f}")
print(f"RMSE : {rmse:,.2f}")
print(f"R²   : {r2:.4f}")

# 10. Actual vs predicted visualization
plt.figure(figsize=(7, 5))
plt.scatter(y_test, y_pred, alpha=0.7)
min_val = min(y_test.min(), y_pred.min())
max_val = max(y_test.max(), y_pred.max())
plt.plot([min_val, max_val], [min_val, max_val], linestyle="--")
plt.xlabel("Actual House Price")
plt.ylabel("Predicted House Price")
plt.title("Actual vs Predicted House Prices")
plt.tight_layout()
plt.show()

# 11. Display six sample predictions
results = pd.DataFrame({
    "Actual Value": y_test.values,
    "Predicted Value": y_pred,
    "Error": y_test.values - y_pred
})
print("\nSample Predictions:\n", results.head(6).round(2))
```

15. Output / Results

Insert or paste only the relevant outputs produced by the program. Leave adequate space for tables, plots, screenshots, or printed metrics.

Dataset shape: (545, 13)
Training samples: 436
Testing samples: 109
Target variable: price
Model: Multiple Linear Regression with one-hot encoding for categorical attributes

Data quality checks:
• Missing values: 0
• Duplicate rows: 0
• Categorical features: encoded using OneHotEncoder(drop="first")

Actual evaluation results from the executed code (80:20 split, random_state = 42):
• MAE = 970,043.40
• MSE = 1,754,318,687,330.66
• RMSE = 1,324,506.96
• R^2 Score = 0.6529

16. Model Evaluation

Report metrics appropriate to the learning task. Delete or leave blank the tables that do not apply.

A. Regression

Metric	Value
MAE	970,043.40
MSE	1,754,318,687,330.66
RMSE	1,324,506.96
R^2 Score	0.6529

B. Classification

Metric	Value
Accuracy	Not applicable for this regression experiment
Precision	Not applicable for this regression experiment
Recall	Not applicable for this regression experiment

F1-Score	Not applicable for this regression experiment
ROC-AUC, if applicable	Not applicable for this regression experiment

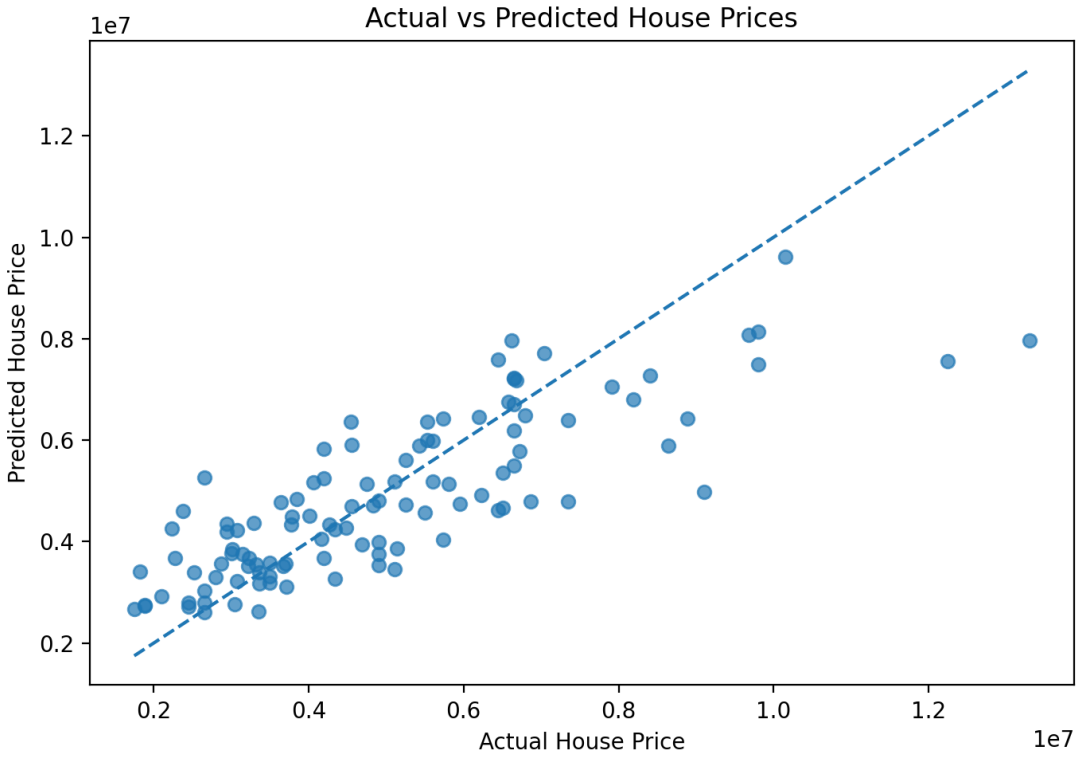
C. Clustering

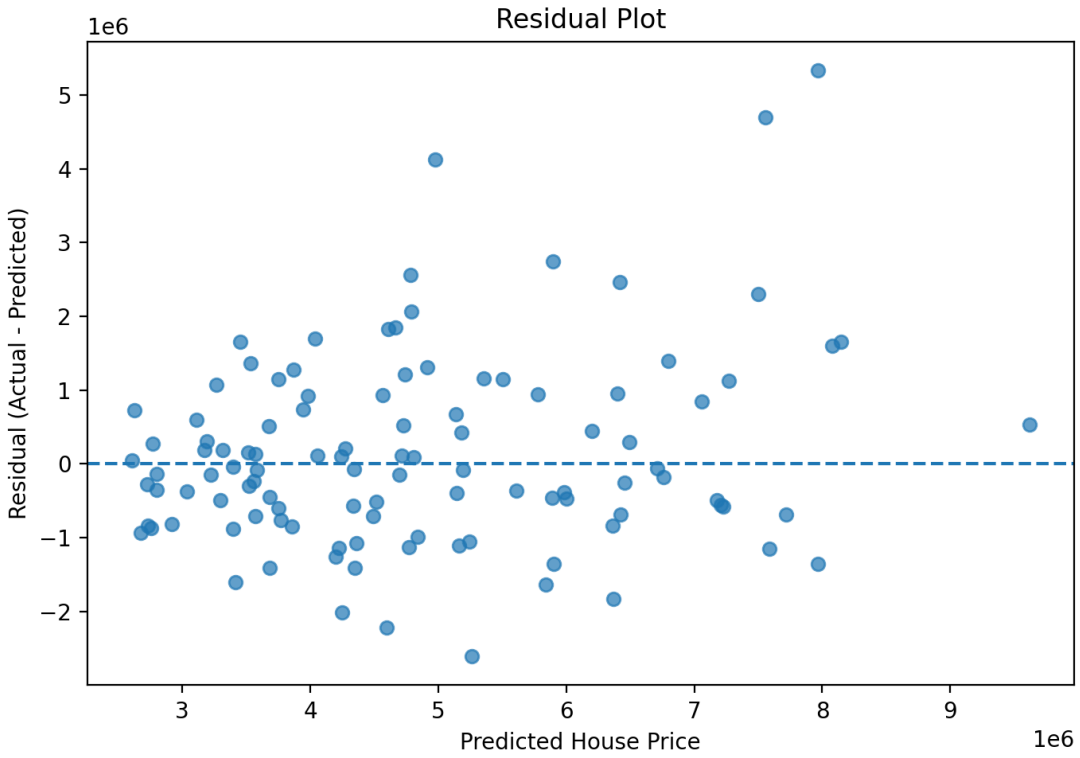
Metric	Value
Silhouette Score	Not applicable for this regression experiment
Davies-Bouldin Index	Not applicable for this regression experiment
Number of Clusters	Not applicable for this regression experiment

17. Visualization of Results

Insert the most relevant visualization(s), such as regression line, actual-vs-predicted plot, confusion matrix, ROC curve, loss curve, decision boundary, cluster plot, or feature-importance graph.

Figure 1: Actual vs Predicted House Prices (generated from the executed Linear Regression model)





Interpretation of Graph / Plot

The graph compares the actual house prices in the test set with the prices predicted by the trained model. Points closer to the diagonal reference line represent accurate predictions. The visible spread around the line shows prediction error; the model achieved an R^2 score of 0.6529, indicating that it explains about 65.29% of the variation in the test-set house prices.

18. Sample Predictions

Sample	Actual Value	Predicted Value	Error / Remark
1	4,060,000	5,164,653.90	-1,104,653.90
2	6,650,000	7,224,722.30	-574,722.30
3	3,710,000	3,109,863.24	600,136.76
4	6,440,000	4,612,075.33	1,827,924.67
5	2,800,000	3,294,646.26	-494,646.26

6	4,900,000	3,532,275.10	1,367,724.90

19. Observations and Analysis

Interpret the result rather than merely repeating the printed output. Comment on model performance, errors, parameter effects, overfitting/underfitting, and any notable patterns.

1. The Kaggle dataset contains 545 records and 12 predictor features; no missing values or duplicate rows were found.

2. Numerical and categorical housing attributes were successfully combined using a preprocessing pipeline with one-hot encoding.

3. The reproducible 80:20 split with random_state = 42 produced 436 training samples and 109 testing samples.

4. The executed model produced MAE = 970,043.40 and RMSE = 1,324,506.96; these values show the typical magnitude of prediction error in the dataset price units.

5. The R^2 score of 0.6529 indicates moderate predictive performance, with approximately 65.29% of the variation in test-set prices explained by the linear model.

6. Larger errors for some houses suggest that multicollinearity, outliers, omitted variables, and non-linear relationships limit a simple Linear Regression model.

20. Result

Write the final outcome in 2–4 sentences. Mention the algorithm, dataset, important performance metric(s), and whether the objective was achieved.

A Multiple Linear Regression model was implemented using the Kaggle Housing Prices Dataset containing 545 house records and 12 predictor features. Using an 80:20 train-test split with random_state = 42, the model achieved MAE = 970,043.40, RMSE = 1,324,506.96, and R^2 = 0.6529. Therefore, the objective of building and evaluating a reproducible housing-price regression model was achieved.

21. Conclusion

Summarize what was learned, whether the objectives were achieved, major limitations, and possible improvements.

This experiment demonstrated the complete workflow of a supervised regression problem using a real Kaggle housing dataset: loading data, checking data quality, preprocessing categorical features, train-test splitting, Multiple Linear Regression training, prediction, evaluation, and visualization. The objectives were achieved, and the measured R^2 score of 0.6529 showed moderate predictive ability. Major limitations include linear-model assumptions, multicollinearity, outliers, and non-linear effects; future work can compare Ridge, Lasso, Random Forest, and Gradient Boosting models.